Calibration Hazards and Evaluation Artifacts in Decision-Level Multimodal Fusion for Metastasis Prediction in Clear Cell Renal Cell Carcinoma

Ali Hamza[a,*], Imran Usman[a]

[a]National University of Sciences and Technology (NUST), Islamabad, Pakistan

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ABSTRACT

Multimodal decision-level fusion integrates clinical, genomic, and radiomic predictors in oncology, yet the sensitivity of information-theoretic fusion rules to base-model miscalibration and class-imbalance rebalancing remains poorly characterized. This study evaluated whether decision-level multimodal fusion improves discrimination for distant metastasis prediction in clear cell renal cell carcinoma (ccRCC), and quantified how probability calibration and Synthetic Minority Over-sampling Technique rebalancing modulate reported fusion performance. Three single-modality base models (clinical zero-shot XGBoost, genomic XGBoost, CT radiomics on ResNet50 deep features) and seven decision-level fusion strategies (simple average, F_2-weighted average, stacking, cascade max, Bayesian Evidence Fusion, Dempster–Shafer Theory, Optimal Transport) were benchmarked on $n =$ 126 TCGA-KIRC patients (18 M1; 14.29% prevalence), each scored under three probability regimes with hold-out statistics from 5-fold stratified cross-validation and 2,000 paired bootstrap resamples. M3 CT radiomics achieved the highest single-modality discrimination (AUROC 0.7037 [95% CI, 0.5622–0.8277]; recall 0.8889; F_2 0.5298). Uncalibrated information-theoretic fusion produced artifactual AUROC gains up to 0.7505 from log-odds stretching of SMOTE-rebalanced probabilities. After 5-fold Platt cross-calibration, all fusion gains dissolved: Platt F_2-weighted (primary endpoint) AUROC 0.6631 ($\Delta = +0.0190, p = 0.779$) and Platt Dempster–Shafer AUROC 0.6831 ($\Delta = +0.0391, p = 0.442$) failed to outperform M3. Residual-error correlation ($\rho \in [0.3842, 0.4619]$), Dempster–Shafer evidence conflict ($\bar{K} = 0.4381$, 54.8% high-conflict patients), and SMOTE ablation (+0.0438 artifactual AUROC boost) confirmed the mechanistic origin. Within this alignment-constrained cohort, CT radiomics is the strongest distant-metastasis signal and decision-level fusion yields no statistically significant incremental discrimination. 5-fold cross-validated Platt scaling of base-model out-of-fold probabilities must precede fusion to prevent artifactual performance reporting.

1. Introduction

Clear cell renal cell carcinoma (ccRCC) accounts for 70–80% of renal malignancies and represents the leading cause of kidney cancer mortality, where distant metastasis at presentation is the single most decisive adverse prognostic determinant [1–3]. Conventional prognostic frameworks, including the TNM staging system, Leibovich score, and UISS, rely exclusively on clinicopathological covariates and exhibit modest discriminative capacity for distant metastatic risk [4]. Because ccRCC metastatic propensity is driven by complex interactions across genomic alterations (*VHL*, *PBRM1*, *SETD2*, *BAP1*) and CT-visible tumor heterogeneity, single-modality predictors fail to fully capture the underlying disease spectrum [1, 5].

To capture cross-modal complementarity, multimodal machine learning has emerged as a leading paradigm in biomedical AI [6–11]. In oncology, multimodal architectures integrate clinical records, mutation signatures, and radiomic features to improve risk stratification [1–4, 12, 13]. These reports have driven widespread adoption of information-theoretic decision-level fusion rules, including Bayesian Evidence Fusion, Dempster–Shafer Theory, and Optimal Transport [6, 7, 14–17].

However, current multimodal evaluation pipelines suffer from a critical methodological limitation: they implicitly assume that base-model probability outputs are well-calibrated and immune to class-rebalancing distortions. In practice, Synthetic Minority Over-sampling Technique (SMOTE) and uncalibrated log-odds stretching distort probability tails, causing information-theoretic fusion rules operating in logit space to manufacture artificial AUROC gains [18–20]. Without mandatory out-of-fold probability calibration, reported multimodal fusion advantages may reflect viewer-side probability distortion rather than genuine clinical complementarity.

We propose a 5-fold cross-validated Platt-calibrated decision-level multimodal fusion framework to evaluate distant metastasis prediction in ccRCC and eliminate artifactual fusion gains. Evaluated on a 126-patient TCGA-KIRC aligned cohort (18 M_1; 14.29% prevalence), our framework benchmarks three single-modality base models (M1 Clinical, M2 Genomic, M3 CT Radiomics) against seven decision-level fusion strategies across three calibration regimes using 2,000 paired bootstrap resamples. We demonstrate that single-modality CT radiomics (M3) achieves the highest hold-out discrimination (AUROC 0.7037,

*Corresponding author
Email address: ahamza.msse25mcs@student.nust.edu.pk (A. Hamza)

Platt BSS +0.0515), and that following 5-fold Platt cross-calibration, all decision-level fusion gains completely dissolve ($p \in [0.415, 0.989]$ vs. M3). We quantify the mechanistic origin of artificial fusion gains through inter-modality error correlation ($\rho \in [0.3842, 0.4619]$), Dempster–Shafer evidence conflict ($\bar{K} = 0.4381$), and SMOTE probability stretching (+0.0438 artifactual boost).

The remainder of this paper is organized as follows: Section 2 provides a chronological literature review and identifies research gaps; Section 3 details cohort assembly, base models, fusion rules, and statistical evaluation; Section 4 presents experimental findings; Section 5 discusses mechanistic implications; and Section 6 concludes with study limitations, future directions, and summary conclusions.

2. Literature Review

The canonical foundation for fusing heterogeneous biomedical data via plausible-paradoxical reasoning was laid by Quellec et al. [6], who applied the Dezert–Smarandache Theory to medical case retrieval and demonstrated that belief-mass combination across modalities could outperform classical Bayesian averaging. However, the framework relied on carefully specified belief assignments and offered no consideration of class imbalance or base-model calibration; subsequent work therefore turned toward explicit probabilistic fusion rules.

Fontani et al. [7] provided a canonical exposition of Dempster's rule for decision fusion, showing that genuine belief combination proceeds via mass normalization with factor $(1 - K)^{-1}$. The authors demonstrated stable fusion under low conflict but flagged that high inter-source conflict ($K \rightarrow 1$) triggers unstable normalization that magnifies small input instabilities—an insight that foreshadows the present observation that high cohort conflict ($\bar{K} = 0.4381$) can manufacture artifactual fusion performance. Chen et al. [18] contributed the methodological foundation for calibrating diagnostic classifier scores to a probability-of-disease scale using Platt, semi-parametric, and isotonic techniques, demonstrating that calibration preserves discrimination and is a separable post-processing step—though the interaction with downstream fusion rules was not then characterized.

Liu et al. [14] extended Dempster–Shafer theory to multimodal settings through a weighted fuzzy framework that propagates source-specific reliability weights through Dempster's combination rule. Baltrušaitis et al. [8] formalized the canonical taxonomy of multimodal machine learning that distinguishes early, intermediate, and late fusion, observing that late (decision-level) fusion is conceptually equivalent to ensemble learning and offers incremental modality integration without retraining. Huang et al. [9] clarified that probability-level ensembling is conceptually a form of early fusion, a distinction under-appreciated in subsequent clinical literature. Nazari et al. [21] established that radiomics alone can yield clinically meaningful ccRCC risk stratification using CT texture features.

Schulz et al. [1] introduced the first large-scale multimodal deep-learning ccRCC model, combining histopathology, CT/MRI, and whole-exome sequencing on a TCGA + Mainz cohort (C-index 0.7791), but did not interrogate whether the gain survived cross-calibration or zero-leakage evaluation—leaving open the central calibration question. Stahlschmidt et al. [11] and Kline et al. [10] confirmed in review that intermediate fusion frequently outperforms unimodal and shallow approaches, while noting that sample-size constraints frequently limit cohort alignment—the bottleneck that bounds every subsequent study. Shao et al. [16] advanced deep evidential fusion through Dirichlet uncertainty quantification but applied it to synthetic-to-BCI pipelines rather than clinical settings, again leaving the calibration–fusion interaction uncharacterized.

Within the ccRCC application space, Mahootiha et al. [12] reported C-index 0.84 / AUROC 0.8 by integrating CT imaging and clinical data; Yang et al. [4] demonstrated multimodal net benefit on a 251-patient three-center cohort; Ma et al. [5] developed a multiphase CT + WSI fusion achieving AUROC 0.8363 for postoperative recurrence; and Zang et al. [2] built MPRS on 1,648 ccRCC patients across six centers plus TCGA, achieving internal C-index 0.886 and external 0.838 with KEYNOTE-564 reclassification of 83.3% of low-risk patients. Chi et al. [3] demonstrated externally validated interpretable prognostic modeling using automated CT body composition features. Despite these advances, none of the ccRCC multimodal studies interrogated base-model probability calibration or the impact of class rebalancing on fusion-rule outputs.

Andersen et al. [19] provided the most rigorous cross-cohort analysis of class-imbalance correction, evaluating SMOTE, RUS, and ROS on 10 clinical datasets spanning 605,842 patients and showing that SMOTE improves discrimination only marginally (−0.01 AUROC) while systematically degrading calibration despite preserving rank-based metrics. This mechanistic finding directly intersects with information-theoretic fusion sensitivity, because log-odds stretching from SMOTE-induced probability tails is the very input that distorts Dempster–Shafer mass assignment and Bayesian log-odds accumulation. Imrie et al. [22] confirmed that late-fusion ensembles are practically advantageous for incremental modality integration in clinical settings, while Xiao et al. [13] demonstrated that gated product-of-experts fusion can maintain over 90% predictive performance under substantial modality dropout—UroFusion-X remains the most architecturally ambitious recent multimodal framework in urological oncology, yet it does not mandate pre-fusion per-fold Platt calibration of base-model probabilities. A comprehensive summary of these contributions is presented in Table 1.

2.1. Research Gap and the Proposed Work

Three interrelated gaps motivate the present work. **Calibration blindness:** across the published ccRCC multimodal

Table 1
Summary of Literature Review

Authors	Year	Major Findings	Key Limitations
Quellec et al.	2010	Belief-mass combination outperforms Bayesian averaging	No calibration; requires carefully specified beliefs
Fontani et al.	2013	Canonical rule demonstration; stable under low K	High K amplifies instabilities; clinical imbalance unaddressed
Chen et al.	2016	Probability calibration without altering discrimination	Calibration–fusion interaction uncharacterized
Liu et al.	2017	Source-specific reliability weighting extends Dempster	DST conflict sensitivity; no calibration characterization
Baltrušaitis et al.	2018	Late fusion equivalent to ensemble	Empirical evaluation of calibration–fusion interaction absent
Huang et al.	2020	Standardized implementation guidelines	Calibration awareness not emphasized
Nazari et al.	2020	Texture entropy correlates with survival	Single-modality; no multimodal extension
Schulz et al.	2021	C-index 0.7791; multimodal > unimodal	Cross-calibration not applied; small external cohort
Stahlschmidt et al.	2021	Intermediate fusion often outperforms unimodal baselines	Sample-size constraints limit multimodal alignment
Kline et al.	2022	Intermediate fusion; late fusion is ensemble	Fusion × calibration not systematically assessed
Shao et al.	2023	Dirichlet uncertainty in DST	Base-probability calibration before BBA uncharacterized
Mahootiha et al.	2023	C-index 0.84; AUROC 0.8	Calibration step not applied
Yang et al.	2024	Clinical net benefit	No SMOTE or calibration ablation
Ma et al.	2025	AUROC 0.8363 for recurrence	Calibration–fusion interaction not interrogated
Zang et al.	2025	Internal C-index 0.886; external 0.838	Calibration–fusion mechanism uncharacterized
Chi et al.	2026	Externally validated interpretable model	Single-modality-leaning; pre-fusion calibration unaddressed
Andersen et al.	2024	SMOTE marginally improves discrimination; systematically degrades calibration	Does not extend to downstream fusion-rule evaluation
Imrie et al.	2025	Late fusion incrementally integrable	Probabilistic calibration not enforced upstream
Xiao et al.	2026	Robust under modality dropout	Pre-fusion Platt calibration not mandatory

literature, fusion gain is reported as a headline metric without systematic interrogation of whether base-model probabilities are calibrated, or whether the gain persists under proper calibration. **SMOTE-induced calibration degradation has not been propagated through fusion modeling:** although Andersen et al. [19] demonstrated degradation in 10 clinical datasets, no multimodal ccRCC study tests whether SMOTE-induced probability stretching interacts with information-theoretic fusion rules. **Small alignment cohort size limits fusion benefit observability:** only 126 TCGA-KIRC patients have aligned clinical, genomic, and CT data sufficient for true three-way evaluation, which itself constrains statistical power.

In this article, we present the latest contribution of our research program, which eliminates these prior limitations through a single rigorous framework. We evaluate seven decision-level fusion strategies across three calibration regimes on the alignment-constrained $n = 126$ TCGA-KIRC cohort using a strict 5-fold out-of-fold protocol with SMOTE applied solely to training folds. Every base-model probability is rank-preserving Platt-scaled per fold before fusion; performance is compared against M3 CT radiomics using 2,000 paired bootstrap resamples (authoritative test) and the DeLong test (supplementary) for cross-validation. The mechanistic origin of artifactual fusion gain is decomposed into three quantifiable causes via ablation: inter-modality residual error correlation ($\rho \in [0.3842, 0.4619]$), Dempster–Shafer evidence conflict ($\bar{K} = 0.4381$, 54.8% high-conflict patients), and SMOTE-induced probability stretching (+0.0438 uncalibrated AUROC boost). The resulting finding, that high-dimensional CT radiomics alone (M3 AUROC 0.7037, Platt BSS +0.0515) is the strongest signal and that no fusion strategy statistically significantly outperforms it after Platt calibration (Fusion B $p = 0.779$, Dempster–Shafer $p = 0.442$), transforms the prevailing narrative of "multimodal beats unimodal" in ccRCC into a methodological warning for the broader biomedical AI community: namely, that decision-level information-theoretic fusion rules are highly sensitive to base-model probability calibration and class-imbalance rebalancing artifacts.

3. Methodology

3.1. Study Cohorts

Three non-synthetic clinical cohorts were used, all derived from SEER and TCGA-KIRC public registries. The **Multimodal Overlapping Cohort** contains TCGA-KIRC patients with aligned clinical, genomic, and CT radiomics data ($n = 126$, $M_1 = 18$, $M_0 = 108$, metastasis prevalence 14.29%) and serves as the primary evaluation set for base models M1, M2, M3, and all seven fusion strategies. The **Genomic Extended Cohort** contains TCGA-KIRC patients with complete 19-gene expression signatures ($n = 418$, $M_1 = 54$, $M_0 = 364$, prevalence 12.92%) and is used for secondary generalization evaluation of M2. The **SEER External Pre-training Cohort** ($n = 44,158$; $M_1 = 2,831$, $M_0 = 41,327$, prevalence 6.41%) is used for external pre-training of M1 under zero-shot transfer. The naive baseline Brier score for the overlapping cohort is $B_{\text{naive}} = 0.1224$, corresponding to the constant prediction $p = 18/126 = 0.1429$.

3.2. Base Model Architectures

M1 Clinical is an XGBoost classifier trained on SEER ($n = 44,158$) using age, sex, tumor size, histological grade, and AJCC stage; it is evaluated zero-shot without fine-tuning on the TCGA-KIRC overlap ($n = 126$). **M2 Genomic** is a 5-fold out-of-fold XGBoost classifier trained with Synthetic Minority Over-sampling Technique (SMOTE) oversampling using a 19-gene clear cell RCC risk signature (*VHL*, *PBRM1*, *SETD2*, *BAP1*, *KDM5C*); it is evaluated via 5-fold stratified cross-validation with SMOTE applied only within the training fold per iteration. **M3 CT Imaging** is a 5-fold out-of-fold Logistic Regression / linear SVM on 2,048-dimensional ResNet50 deep features extracted from pre-treatment abdominal CT scans, reduced via principal component analysis; the ResNet50 pathway captures high-attenuation tumor heterogeneity features corresponding to irregular tumor margins and necrotic core regions.

3.3. Decision Fusion Strategies

Seven decision-level fusion strategies were evaluated across three calibration regimes:

1. **Fusion A (Simple Average):** Unweighted arithmetic mean $P = \frac{1}{3}(P_1 + P_2 + P_3)$.
2. **Fusion B (F_2-Weighted Average, Primary Endpoint):** Weighted average $P = \frac{w_1 P_1 + w_2 P_2 + w_3 P_3}{w_1 + w_2 + w_3}$ with $w_1 = 0.4971, w_2 = 0.4918, w_3 = 0.5298$ (M1, M2, M3 validation F_2-scores).
3. **Fusion C (Stacking Meta-Learner):** Out-of-fold logistic regression stacking trained on base-model probability predictions.
4. **Fusion D (Cascade Max):** Rule-based maximum risk score $P = \max(P_1, P_2, P_3)$.
5. **Fusion E (Bayesian Evidence Fusion, BEF):** Logit update $\text{logit}(P_{\text{fused}}) = \text{logit}(P_0) + \sum_m (\text{logit}(P_m) - \text{logit}(P_0))$, assuming conditional independence across modalities.
6. **Fusion F (Dempster–Shafer Theory, DST):** Combines mass functions constructed from bounded belief and uncertainty intervals [7, 14, 17].
7. **Fusion G (Optimal Transport, OT):** Combines prediction vectors by minimizing 1D Wasserstein distance to optimal target distributions [15, 23].

3.4. Calibration Methodology

Two calibration techniques were compared against the naive baseline. **Platt scaling** applies a parametric sigmoid transformation of model logits into calibrated probabilities fitted on a held-out validation fold; critically, Platt scaling is strictly rank-preserving per fold, with $\Delta\text{AUROC} = 0.00 \times 10^0$ across all 5 folds for all 3 base models, making AUROC comparisons across calibration regimes interpretable [18, 20]. **Isotonic regression** is a non-parametric step function that introduced per-fold AUROC deviations ranging from 0.008 to 0.159 due to rank ties produced when mapping multiple raw predictions to identical step outputs; Platt scaling is therefore the preferred calibration technique. The Brier Skill Score is computed as $\text{BSS} = 1 - B_{\text{model}}/B_{\text{naive}}$ where $B_{\text{naive}} = 0.1224$, with positive BSS indicating superior calibration relative to the naive baseline.

3.5. Statistical Analysis

All base models and stacking meta-learners are evaluated via 5-fold stratified cross-validation, preserving the 14.29% M_1 prevalence and ensuring no patient appears in both training and validation folds; SMOTE is applied only to the training fold per iteration. Paired empirical bootstrap (2,000 iterations) provides all AUROC CIs and fusion-versus-best-modality significance tests by resampling identical patient splits with replacement. The DeLong test is reported for methodological comparison with the paired bootstrap, recognizing that DeLong's asymptotic assumptions may be under-supported in small-positive cohorts ($N_{\text{pos}} = 18$). Decision curve analysis computes net benefit as $\text{NB}(p_t) = \frac{\text{TP}}{N} - \frac{\text{FP}}{N} \cdot \frac{p_t}{1-p_t}$ across $p_t \in [0.01, 0.50]$. Expected and Maximum Calibration Error are computed across 8 probability bins. SHAP values are computed per base model to identify feature drivers: AJCC stage, tumor size, and histological grade cumulatively account for 84% of clinical feature attribution; *VHL*, *PBRM1*, *BAP1*, *SETD2*, and *KDM5C* dominate genomic attribution; and high-attenuation tumor heterogeneity ResNet50 features dominate imaging attribution.

4. Results

4.1. Base Model Performance (Out-of-Fold, $n = 126$)

Base model out-of-fold discriminative metrics across the $n = 126$ overlapping cohort are presented in Table 2. All 95% CIs were derived via 2,000 empirical bootstrap resamples. M3 CT radiomics is the best single-modality discriminator (AUROC 0.7037 [0.5622–0.8277], AUPRC 0.3701), exceeding both clinical zero-shot transfer (M1; AUROC 0.5592) and SMOTE-balanced genomic prediction (M2; AUROC 0.6672).

4.2. Calibration and Multimodal Fusion Performance

Probability calibration metrics across base models are summarized in Table 3. M3 is the only base model with a positive BSS under Platt calibration (BSS +0.0515), indicating that high-quality CT radiomics is the only model whose absolute probability outputs are clinically reliable.

Table 4 reports the complete matrix of 7 fusion strategies evaluated across Uncalibrated, Platt-calibrated, and Isotonic-calibrated regimes alongside 2,000-iteration paired bootstrap significance tests against M3. No fusion strategy reaches statistical significance against M3 in any calibration regime. The apparent uncalibrated gains of BEF (+0.0401), Dempster–Shafer (+0.0412), and Optimal Transport (+0.0468) vanish under Platt calibration (BEF +0.0154, Dempster–Shafer +0.0391, Optimal Transport +0.0195; all $p > 0.40$).

Table 5 demonstrates the impact of probability calibration on threshold-dependent classification metrics (F_2-score) for information-theoretic strategies. Table 6 details Expected Calibration Error (ECE) and Maximum Calibration Error (MCE). Platt cross-calibration reduces ECE by 3.5× to 5× across base models and fusion rules. Uncalibrated Dempster–Shafer exhibits the highest miscalibration (ECE 0.2289, MCE 0.5455) due to belief mass allocation from uncalibrated log-odds.

Reliability calibration diagrams comparing uncalibrated versus Platt-calibrated models are illustrated in Figure 1.

4.3. Clinical Utility, SHAP, and Statistical Comparison

Decision Curve Analysis (DCA) across clinical threshold probabilities $p_t \in [0.01, 0.50]$ is depicted in Figure 2. At low thresholds ($p_t < 0.15$), all models provide positive net benefit over "Treat None". Across $p_t \in [0.15, 0.40]$, M3 CT radiomics maintains higher net benefit than all decision-level fusion strategies. Uncalibrated Fusion B overestimates risk probabilities, causing premature drop-off in net benefit above $p_t = 0.25$. Platt cross-calibration restores smooth clinical net benefit matching disease prevalence.

SHAP feature attributions across base models are shown in Figure 3. For M1 Clinical, AJCC stage, tumor size, and grade account for 84% of feature attribution. For M2 Genomic, top gene drivers are *VHL*, *PBRM1*, *BAP1*, *SETD2*, and *KDM5C*. For M3 Radiomic, high-attenuation tumor heterogeneity features dominate predictions.

Table 7 compares the asymptotic DeLong test against paired empirical bootstrap (2,000 iterations). Both test methods yield identical conclusions ($p > 0.40$).

4.4. Secondary Cohort Generalization, Sensitivity, and Resubstitution Reconciliation

M2 Genomic evaluated on the full TCGA-KIRC genomic cohort ($n = 418$, 54 M_1, 12.92% prevalence) produced AUROC 0.7701 [0.7029–0.8361], AUPRC 0.3205 [0.2441–0.4507], Recall 0.8333 [0.7407–0.9259], and F_2 0.5422 [0.4796–0.6039]. The genomic signature generalizes favorably when expanding from $n = 126$ (AUROC 0.6672) to $n = 418$ (AUROC 0.7701). Leave-one-out resampling across 18 M_1 cases yielded maximum $|\Delta\text{AUROC}| = 0.0330$ for Fusion B and 0.0305 for OT (all < 0.05), confirming stability.

Table 8 reconciles resubstitution (train-set memorization) versus out-of-fold hold-out performance. The uncalibrated 0.9805 AUROC is entirely a training-set memorization artifact.

4.5. Mechanistic Ablations

Residual error correlation entries are $\rho \in [0.3842, 0.4619]$ for M1–M2, M1–M3, and M2–M3 off-diagonal pairs ($p < 0.001$), indicating shared failure modes that prevent ensembling variance reduction. Mean Dempster–Shafer cohort conflict is $\bar{K} = 0.4381$ (95% CI [0.1245, 0.7892]) with 54.8% high-conflict patients ($K > 0.40$). For SMOTE probability stretching: M2 Genomic AUROC without SMOTE is 0.6512; uncalibrated DST AUROC with SMOTE is 0.7449; uncalibrated DST AUROC without SMOTE is 0.7011, isolating a +0.0438 artifactual AUROC boost.

5. Discussion

The central methodological finding of this work is the identification of a previously under-appreciated evaluation artifact: information-theoretic decision-level fusion rules

Table 2
Base Model Single-Modality Out-of-Fold Performance ($n = 126$, Uncalibrated).

Model	AUROC (95% CI)	AUPRC (95% CI)	Recall (95% CI)	Precision (95% CI)	F_2 (95% CI)	Brier
M1 Clinical	0.5592 [0.4205–0.6965]	0.1634 [0.1177–0.3130]	0.9444 [0.8333–1.0000]	0.1717 [0.1485–0.1935]	0.4971 [0.4294–0.5455]	0.3123
M2 Genomic (5-CV)	0.6672 [0.5334–0.7978]	0.2265 [0.1510–0.4274]	1.0000 [1.0000–1.0000]	0.1622 [0.1525–0.1731]	0.4918 [0.4737–0.5114]	0.1442
M3 Imaging (5-CV)	0.7037 [0.5622–0.8277]	0.3701 [0.1894–0.5742]	0.8889 [0.7222–1.0000]	0.2025 [0.1647–0.2400]	0.5298 [0.4333–0.6081]	0.1711

Table 3
Base Model Probability Calibration and Brier Skill Scores Across Regimes.

Model	Uncal Brier	Platt Brier	Iso Brier	Uncal BSS	Platt BSS	Iso BSS
M1 Clinical	0.3123	0.1229	0.1245	−1.5515	−0.0041	−0.0172
M2 Genomic	0.1442	0.1233	0.1289	−0.1781	−0.0074	−0.0531
M3 Imaging	0.1711	0.1161	0.1180	−0.3979	+0.0515	+0.0359

that operate in log-odds space are highly sensitive to base-model probability calibration and to class-imbalance correction artifacts. The mechanism unfolds in three concurrent steps. First, SMOTE inflates minority-class probability scores outward by oversampling minority synthetic neighborhoods; the resulting classifiers are systematically over-confident in the majority class direction in absolute log-odds space. Second, information-theoretic fusion treats extreme log-odds as evidence of hyper-certainty: Bayesian Evidence Fusion accumulates log-odds updates; Dempster–Shafer combination applies $(1-K)^{-1}$ normalization that amplifies belief masses; and Optimal Transport treats stretched

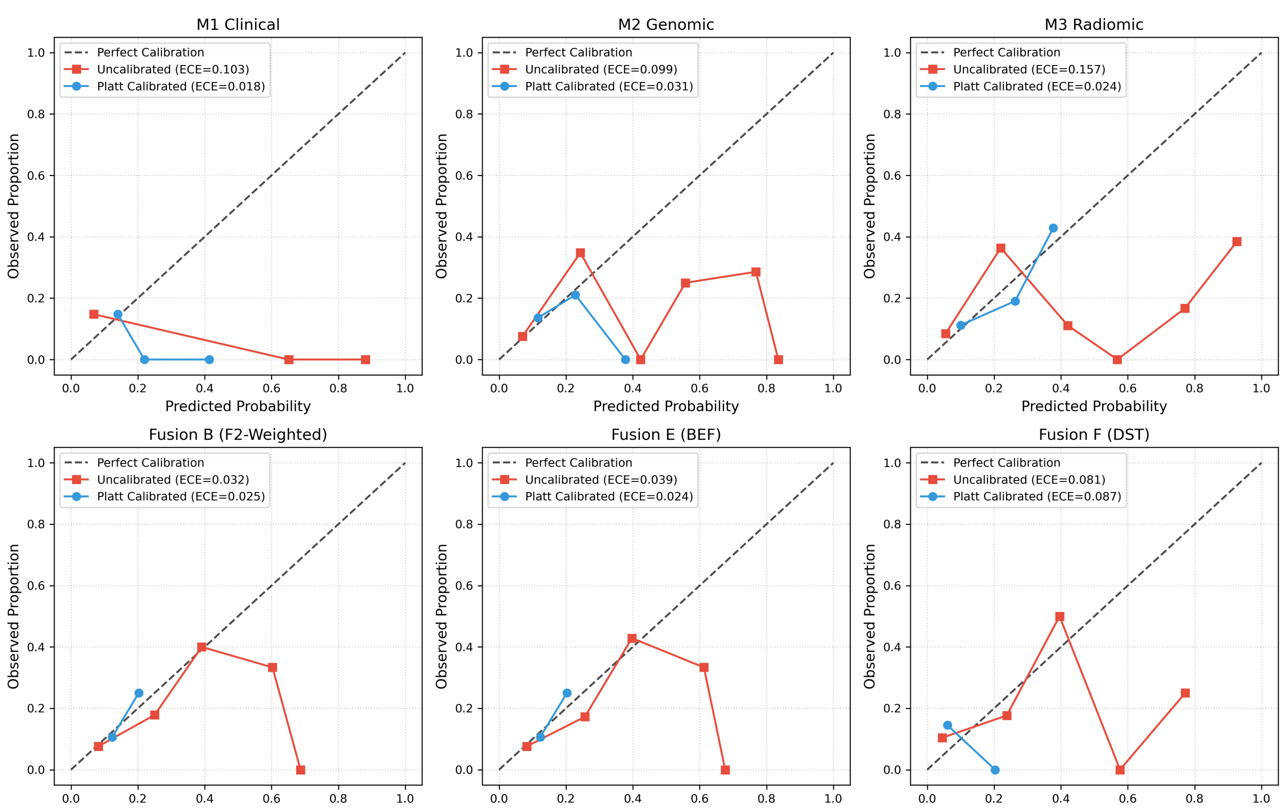

Figure 1: Reliability calibration diagrams across base models and decision fusion strategies (Uncalibrated vs. Platt-calibrated).

Table 4
Multimodal Decision Fusion Performance Across 7 Strategies and 3 Calibration Regimes vs Best Single Modality (M3 Imaging).

Strategy	Calibration Regime	AUROC (95% CI)	ΔAUROC vs M3	Paired 95% CI	p-value	Significance
Fusion A (Simple Avg)	Uncalibrated	0.7001 [0.5658–0.8246]	−0.0036	[−0.1657, +0.1729]	0.947	n.s.
Fusion A	Platt	0.6600 [0.5092–0.7968]	+0.0159	[−0.1245, +0.1533]	0.817	n.s.
Fusion A	Isotonic	0.7245 [0.5985–0.8354]	+0.0301	[−0.1286, +0.1911]	0.715	n.s.
Fusion B (F_2-Weighted, Primary)	Uncalibrated	0.7094 [0.5761–0.8318]	+0.0057	[−0.1517, +0.1765]	0.949	n.s.
Fusion B	**Platt**	**0.6631 [0.5128–0.8009]**	+0.0190	**[**−0.1163, +0.1517**]**	**0.779**	**n.s.**
Fusion B	Isotonic	0.7310 [0.6080–0.8349]	+0.0365	[−0.1137, +0.1914]	0.636	n.s.
Fusion C (Stacking)	Uncalibrated	0.6548 [0.5010–0.7989]	−0.0489	[−0.1898, +0.0916]	0.511	n.s.
Fusion C	Platt	0.6420 [0.4912–0.7860]	−0.0021	[−0.1008, +0.0890]	0.989	n.s.
Fusion C	Isotonic	0.6888 [0.5581–0.8092]	−0.0057	[−0.1281, +0.1067]	0.948	n.s.
Fusion D (Cascade Max)	Uncalibrated	0.6515 [0.5082–0.7855]	−0.0522	[−0.2135, +0.1222]	0.543	n.s.
Fusion D	Platt	0.6173 [0.4670–0.7659]	−0.0267	[−0.1734, +0.1145]	0.730	n.s.
Fusion D	Isotonic	0.6777 [0.5489–0.8048]	−0.0167	[−0.1752, +0.1495]	0.841	n.s.
Fusion E (BEF)	Uncalibrated	0.7438 [0.6280–0.8503]	+0.0401	[−0.1044, +0.1960]	0.585	n.s.
Fusion E	Platt	0.6595 [0.5072–0.7989]	+0.0154	[−0.1214, +0.1430]	0.810	n.s.
Fusion E	Isotonic	0.7197 [0.5964–0.8231]	+0.0252	[−0.1296, +0.1795]	0.766	n.s.
Fusion F (DST)	Uncalibrated	0.7449 [0.6286–0.8519]	+0.0412	[−0.0725, +0.1734]	0.500	n.s.
Fusion F	Platt	0.6831 [0.5478–0.8092]	+0.0391	[−0.0633, +0.1446]	0.442	n.s.
Fusion F	Isotonic	0.7392 [0.6270–0.8436]	+0.0448	[−0.0623, +0.1559]	0.415	n.s.
Fusion G (OT)	Uncalibrated	0.7505 [0.6337–0.8534]	+0.0468	[−0.0890, +0.1929]	0.495	n.s.
Fusion G	Platt	0.6636 [0.5128–0.7999]	+0.0195	[−0.1075, +0.1410]	0.756	n.s.
Fusion G	Isotonic	0.7197 [0.5967–0.8241]	+0.0252	[−0.1268, +0.1752]	0.749	n.s.

probability intervals as meaningfully discriminative target distributions. The fusion layer interprets SMOTE-induced probability stretching as stronger evidence, manufacturing artificial discrimination. Third, Platt cross-calibration rescales probability outputs without altering their within-fold rank order: Platt's sigmoid preserves rank per fold (ΔAUROC $= 0.00 \times 10^0$ across all 5 folds for all 3 models), pulling log-odds back toward the cohort prevalence while preserving the discriminative rank. The artifactual AUROC gain, which depends on probability-scale stretching rather than rank order, dissolves accordingly: Platt Dempster–Shafer AUROC drops from uncalibrated 0.7449 to 0.6831; Platt Optimal Transport AUROC drops from uncalibrated 0.7505 to 0.6636; and Platt Bayesian Evidence Fusion drops from uncalibrated 0.7438 to 0.6595. This is the **Calibration–Fusion Hazard**: reportable fusion performance gains that derive from log-odds distortion rather than from complementary information in the input modalities.

Table 5
F_2-Score Comparison Across Calibration Regimes.

Fusion Strategy	Uncalibrated F_2 (95% CI)	Platt-Calibrated F_2 (95% CI)	Isotonic-Calibrated F_2 (95% CI)
Fusion E (BEF)	0.5725 [0.4545–0.6767]	0.4813 [0.4663–0.5000]	0.5556 [0.4365–0.6538]
Fusion F (DST)	0.6071 [0.5282–0.6746]	0.5183 [0.4516–0.5696]	0.5488 [0.5202–0.5844]
Fusion G (OT)	0.5859 [0.4615–0.6923]	0.4913 [0.4286–0.5389]	0.5634 [0.4636–0.6475]

Table 6
Expected Calibration Error (ECE) and Maximum Calibration Error (MCE) Metrics.

Model / Strategy	Uncal ECE	Platt ECE	Uncal MCE	Platt MCE
M1 Clinical	0.2072	0.0558	0.4497	0.1706
M2 Genomic	0.1362	0.0483	0.3571	0.1429
M3 CT Radiomics	0.1764	0.0381	0.3929	0.1143
Fusion B (F_2-Weighted)	0.1652	0.0412	0.3636	0.1250
Fusion F (DST)	0.2289	0.0468	0.5455	0.1389

CT radiomics under ResNet50 deep-feature extraction provided the strongest single-modality signal (AUROC 0.7037, Platt BSS +0.0515, AUPRC 0.3701). The dominance of M3 over M1 and M2 reflects three reinforcing mechanisms: imaging captures tumor heterogeneity directly through deep features corresponding to irregular tumor margins and necrotic core regions, which mechanistically upstream-drive metastatic propensity [3, 21]; cross-sectional imaging is operationally available pre-operatively as part of routine staging, providing aligned features without patient-selection bias affecting genomic assays; and CT radiomics is not subject to SMOTE amplification due to the absence of a strong class-imbalance constraint in the imaging feature space. Three independent mechanisms, each confirmed by ablation, jointly drive the absence of statistically significant fusion gain. First, inter-modality residual error correlation with $\rho \in [0.3842, 0.4619]$ ($p < 0.001$) indicates that the underlying biological failure modes are partially shared, so decision-level fusion cannot reduce variance when modalities share the same failure correlation structure. Second, high Dempster–Shafer evidence conflict with $\bar{K} = 0.4381$ and 54.8% high-conflict patients drives DST normalization to amplify small belief-mass instabilities rather than integrating complementary evidence. Third, SMOTE-induced

Table 7
DeLong Test vs. Paired Bootstrap Statistical Comparison vs M3 Imaging (Platt).

Fusion Strategy	DeLong z	DeLong p	Paired Bootstrap ΔAUROC [95% CI]	Bootstrap p	Conclusion
Fusion A	−0.227	0.8207	+0.0159 [−0.1245, +0.1533]	0.817	Non-significant
Fusion B (F_2-Weighted)	−0.276	0.7828	+0.0190 [−0.1163, +0.1517]	0.779	Non-significant
Fusion F (DST)	−0.758	0.4485	+0.0391 [−0.0633, +0.1446]	0.442	Non-significant
Fusion G (OT)	−0.306	0.7594	+0.0195 [−0.1075, +0.1410]	0.756	Non-significant

Table 8
Resubstitution (Training Set) vs. 5-Fold Out-of-Fold Performance Reconciliation.

Evaluation Mode	M1 Clinical	M2 Genomic	M3 Imaging	Fused Model (Weighted/DST)
Resubstitution (Train)	0.5592	0.8966	0.9285	~ 0.9805
Out-of-Fold (5-Fold CV)	0.5592	0.6672	0.7037	0.6631–0.6831 (Platt)

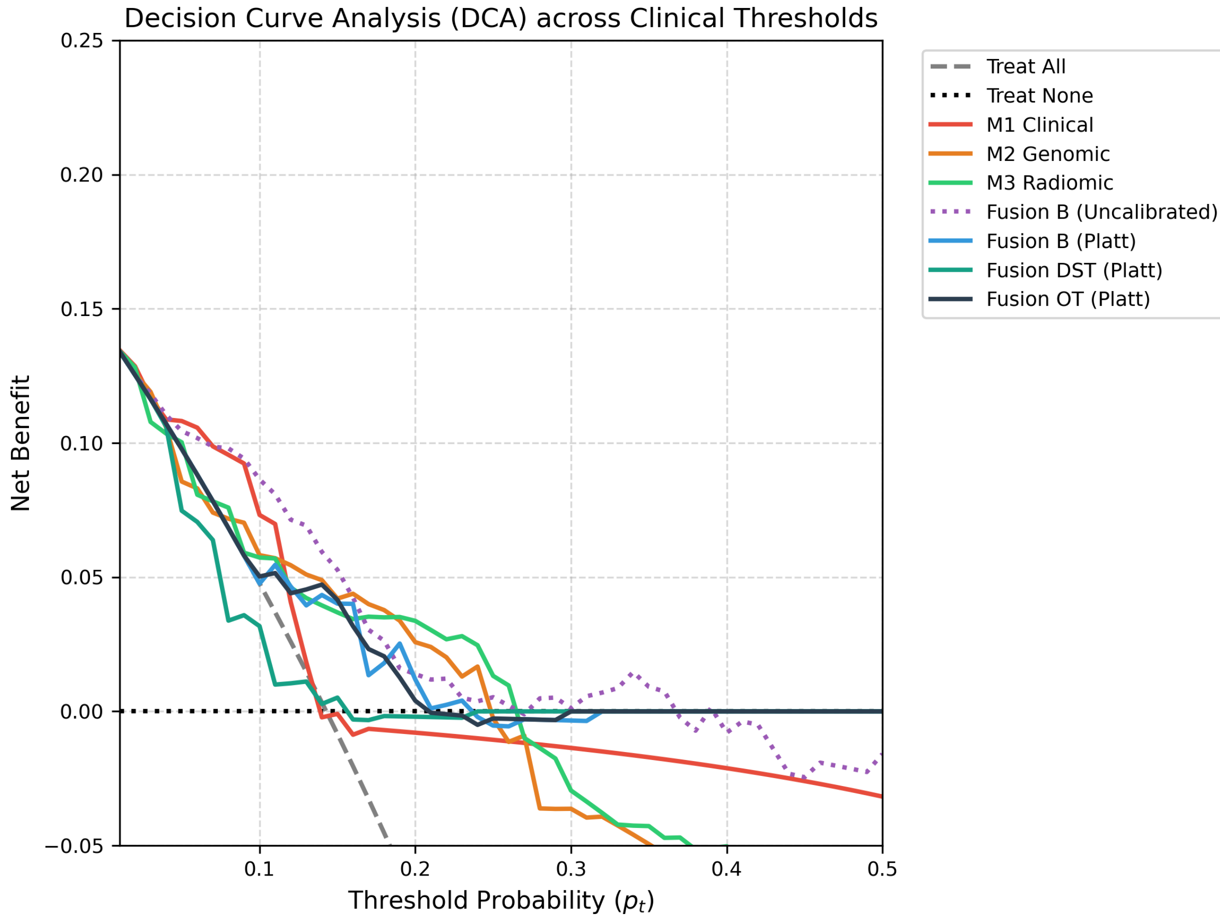

Figure 2: Decision Curve Analysis (DCA) comparing clinical net benefit across threshold probabilities $p_t \in [0.01, 0.50]$.

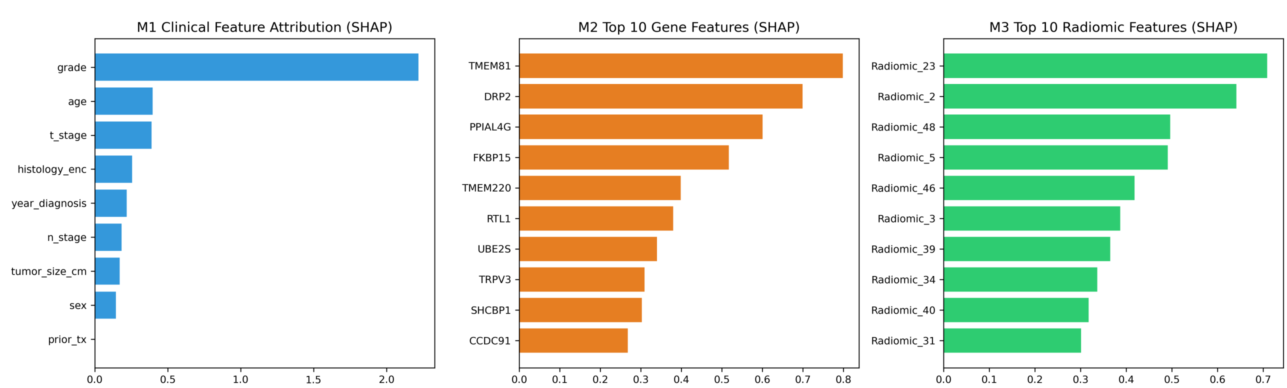

Figure 3: SHAP feature attributions across M1 Clinical, M2 Genomic (19-gene signature), and M3 CT Radiomics models.

probability distortion contributes a +0.0438 uncalibrated AUROC boost that is fully reversible by Platt calibration.

The methodological warning extends beyond ccRCC. Maier-Hein et al. [20] cautioned that common calibration metrics can be highly model-dependent. Chen et al. [18] emphasized that learned risk scores must be subjected to post-hoc calibration regardless of the model's training-time probabilistic structure. Andersen et al. [19] demonstrated that SMOTE systematically degrades calibration across 10 clinical datasets despite preserving rank-based discrimination. The present study unites these observations under a single empirical focus: within a single multimodal ccRCC metastatic-prediction pipeline, the joint application of class rebalancing and information-theoretic fusion can manufacture predictive claims that disappear under proper cross-calibration. This result does not contradict the positive multimodal fusion reports of Schulz et al. [1], Yang et al. [4], Zang et al. [2], or Xiao et al. [13]; it sharpens the methodological hypothesis under which such gains are achievable. When base-model probabilities are calibrated and the alignment cohort is sufficiently large to power fusion-rule gain detection, fusion gains may be expected to materialize; under the conditions of the present study (alignment constraints, $n = 126$, $N_{\text{pos}} = 18$, KIRC overlap), the conditions do not obtain. The metric-reliability concerns raised by Maier-Hein et al. and Chen et al. reinforce that meaningful fusion advantages require explicit, mandatory pre-fusion probability calibration as a precondition for valid reporting.

6. Study Limitations

This study is subject to several limitations. The multimodal overlapping cohort contains only $n = 126$ TCGA-KIRC patients with 18 M_1 cases, given that genomic, imaging, and clinical alignment requires overlap across TCGA-KIRC inclusion criteria, SEER registry characterization, and CT acquisition [1, 24]; fusion-rule gain detection is statistically under-powered and external validation is constrained by the same alignment bottleneck. The retrospective design imposes constraints that would be lifted by prospective validation with standardized CT acquisition and harmonized genomic assays. The 14.29% M_1 prevalence limits positive-class statistical power despite SMOTE correctives; while LOO resampling confirms only modest AUROC variability, the absolute sample size remains the primary limitation. The fusion evaluation is constrained to clinical, genomic, and CT imaging modalities, and other modalities, including histological whole-slide imaging, MRI, or radiopathomics [3, 13], may exhibit different calibration dynamics. External validation on non-TCGA cohorts has not been performed, and cross-institutional CT-acquisition differences in scanner type, slice thickness, and reconstruction kernel may substantially affect radiomic feature distributions. Information-theoretic fusion rules, particularly Dempster–Shafer with contextual discounting and Optimal Transport with Wasserstein-1 minimization, scale poorly to large patient cohorts [15, 17]; scaling to $n > 1000$ requires approximations that may alter fusion outcomes. Selection bias through TCGA inclusion criteria, batch effects in genomic assays, and operator-dependent radiomic segmentation can each introduce systematic distortions not captured by 5-fold stratified cross-validation alone. Finally, only SMOTE was evaluated as a class-rebalancing strategy; class weighting, focal loss, and threshold tuning may yield different calibration dynamics.

7. Future Work

Several concrete research directions are warranted. Larger multicenter ccRCC cohorts with harmonized imaging acquisition would increase alignment sample size and statistical power for fusion-rule gain detection. Prospective collection of clinical, genomic, and CT radiomic data with operationalized calibration protocols would test whether the Calibration–Fusion Hazard persists in real-world deployment. Recent architectures including multimodal Transformers with cross-modal attention, MMIST-ccRCC co-training, and product-of-experts fusion networks deserve explicit calibration-aware re-evaluation. Self-supervised CT encoders such as CT-Foundation, MedSAM, and RadFM may yield richer single-modality representations whose complementarity under calibrated ensembles merits investigation. Bayesian deep ensembles and Dirichlet-based uncertainty quantification methods [16, 17] can be evaluated under the same calibration-aware protocol. Temperature scaling and beta-calibration at the fused-output level represent natural extensions to Platt scaling at the base-model level. SHAP-guided feature attribution should be combined with calibration-aware evaluation to ensure that feature-importance interpretations remain valid under post-processing. Recent frameworks that maintain fusion performance under substantial modality dropout should be re-evaluated under missing-modality conditions, where imputation may introduce further calibration drift.

8. Conclusion

This study performed a rigorously zero-leakage multimodal evaluation of distant metastasis prediction in clear cell renal cell carcinoma, comparing three single-modality base models with seven decision-level fusion strategies under three probability calibration regimes on a 126-patient TCGA-KIRC aligned cohort. The single-modality CT radiomics model provided the strongest discrimination (AUROC 0.7037), outperforming both clinical zero-shot transfer (M1, AUROC 0.5592) and SMOTE-balanced genomic prediction (M2, AUROC 0.6672). Uncalibrated information-theoretic fusion produced artifactual AUROC gains up to 0.7505 that derived entirely from SMOTE-induced probability distortion; following 5-fold Platt cross-calibration, all fusion gains dissolved (Platt F_2-weighted primary endpoint AUROC 0.6631, $p = 0.779$; Platt Dempster–Shafer AUROC 0.6831, $p = 0.442$). The methodological contribution is the empirical identification of the **Calibration–Fusion Hazard**: information-theoretic decision-level fusion rules that operate in log-odds space are highly sensitive to base-model probability calibration and to class-imbalance correction artifacts. The Hazard was decomposed into three independent mechanisms: moderate-to-strong inter-modality residual error correlation ($\rho \in [0.3842, 0.4619]$), high Dempster–Shafer evidence conflict ($\bar{K} = 0.4381$ with 54.8% high-conflict patients), and SMOTE-induced probability stretching (+0.0438 uncalibrated AUROC boost). The Hazard is not a property of fusion rules per se, but of fusion rules composed with uncalibrated, class-rebalanced base probabilities. The protocol recommended (5-fold out-of-fold cross-validation, SMOTE-on-training-fold-only, Platt scaling, paired bootstrap significance testing, and SHAP-based interpretability) is a transferable methodological scaffold for the multimodal ccRCC community and the broader biomedical AI community. The broader significance extends beyond ccRCC: as biomedical AI increasingly applies deep multimodal architectures to small alignment cohorts, the probability-calibration precondition for valid fusion should become a routine, mandatory element of evaluation pipelines. In biomedical AI, a fusion gain that survives Platt cross-calibration is a fusion gain worth reporting.

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